

EAPD7111

ICE task 1

Introduction

In the world of Enterprise Software, we don't reinvent the wheel—we use wheels that have been battle-tested by thousands of engineers before us. These "wheels" are called Design Patterns. Your task is to select one specific pattern and deconstruct it so that fellow developers can benefit from your understanding.

Design patterns to choose from (choose **one**):

- Creational
- Structural
- Behavioral
- Concurrency

1. Draw and label a clear diagram of your chosen pattern.

- Must include: The main classes/objects involved and the relationships between them (e.g., Inheritance, Composition, Implementation).
- Use arrows and labels to show the flow of data or control. How does Object A trigger a change in Object B?

[20]

2. Write 3–5 key points explaining how your chosen pattern is distinct from its "closest relatives."

- *Example:* "While both provide a wrapper, a Decorator adds new responsibilities to an object, whereas an Adapter simply changes its interface to make it compatible with another."

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3. At least 5 Advantages and Disadvantages

Create a simple table or list outline:

- Advantages: (e.g., Decoupling, Reusability, Scalability, etc.).
- Disadvantages: (e.g., Increased code complexity, memory overhead, "over-engineering", etc.).

[10]

[Total = 38]